

Unit

1

Lesson

2

Spark The Challenge!

Discovering Wonders of The Eye.

Time Frame: [90 min].

Ready, Set, Learn!

By the end of this lesson students will be able to:

- Describe the parts of the human eye and explain how they work together to create vision.
- Explain what causes vision loss, including eye injuries, diseases, and conditions like color blindness.
- Interpret labelled diagrams of the eye and describe the function of each part.
- Compare access to eye care and medical support in different parts of the world.
- Understand how color blindness affects the way some people see colors, and what causes it.

Challenge questions of the lesson

1. How do the parts of the eye work together to help us see the world?
2. What happens when one part of the eye doesn't work properly?
3. How do people with visual impairments adapt to daily life?
4. How can we use technology and creativity to help people with vision problems?
5. Is it fair that not everyone in the world has access to eye care?

SEL Relation

- **Empathy:** Students reflect on what it's like to live with visual impairments and simulate these experiences to build deeper understanding.
- **Perspective-Taking:** They explore global differences in access to vision care and learn to appreciate challenges others may face.
- **Social Awareness:** Through group discussions and partner tasks, students develop listening and collaboration skills.
- **Responsible Decision-Making:** Students consider ethical questions around health access and design technologies with others' needs in mind.
- **Self-Awareness:** Students articulate how they learn best—through visual models, diagrams, games, or real-world simulations.

Engage



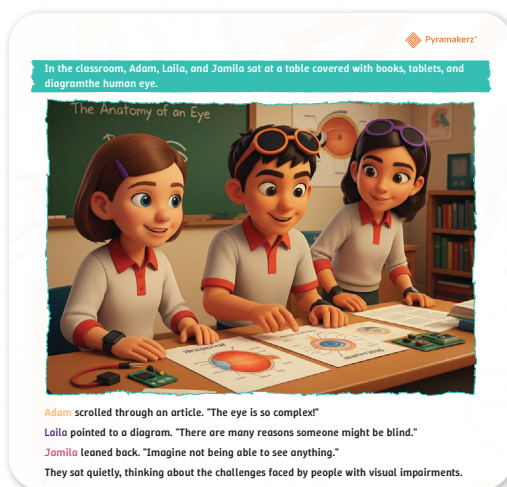
Teacher Role:

- Spark curiosity using visuals or simulations of how the world looks with visual impairment.
- Scaffold key terms like “visual impairment,” “diagram,” and “lens” during discussion.
- Guide reflection and model empathetic thinking through prompts and think-alouds.
- Encourage all responses, reinforcing that each perspective matters.
- Connect student insights back to the real-world challenges of visual impairment.

Expected from students:

- Actively observe story illustrations and listen to dialogue.
- Engage in pair or class discussions using imagination and empathy.
- Reflect on how visual impairments could affect daily life.
- Ask questions or share ideas about how vision helps us understand the world.

Parts of the books included.



Story



1. Start with a Warm Welcome and Story Recap:

- “Good morning, innovators! Last session, we met our friends Adam, Laila, and Jamila. They were exploring how the human eye works. Can anyone remember what they were doing at the table?”
- (Encourage students to recall elements like books, tablets, or eye diagrams. Prompt connections to the idea of vision and its importance.)

2. Continue the Story with Expression:

- “Today, something made them stop and think. Adam found something surprising in an article. Laila noticed a diagram about blindness. Jamila started imagining a life without sight... Let’s listen closely to what happened in their classroom.”
- (Read the full scene aloud using character voices. Pause briefly after Jamila’s line to allow for student reflection.)

3. Follow-Up Questions to Spark Curiosity:

- “How would you feel if you couldn’t see anything around you?”
- “What questions do you have about how blind people understand the world?”
- “Can we think of ways technology might help someone like Mr. Evans, who can’t see?”

Possible Strategies for Implementation:

1. Think–Pair–Share.

- How to Implement:
 - After reading the story aloud, ask: “Why do you think Jamila said ‘Imagine not being able to see anything’?”
 - Let students first think quietly about the question.
 - Then have them pair up with a classmate to discuss their ideas.
 - Finally, share out with the whole class to gather diverse perspectives on how blindness might affect daily life.

2. Picture Walk with Inference.

- How to Implement:
 - Show the story image before reading the text.
 - Ask: “What do you think the characters are doing? What might they be learning?”
 - Encourage students to observe body language, objects (books, diagrams), and expressions before confirming with the story dialogue.
 - Use this strategy to help students make predictions and build visual literacy.

3. Role-Play Snippets

- How to Implement:
 - After reading, assign small roles to three students to act out the dialogue of Adam, Laila, and Jamila.
 - Ask the class to listen for emotion and meaning—What is each character feeling?
 - Follow up with: “Why do you think they reacted that way?” to encourage empathy and perspective-taking.

4. Emotion Mapping

- How to Implement:
 - Use emojis or emotion cards and ask: “How did each character feel in this scene?”

- Let students hold up cards or draw the emoji that matches what they think Adam, Laila, or Jamila felt.
- Ask them to justify their choices using clues from the text (e.g., “Jamila leaned back...”).

5. K-W-L Anchor Chart.

- How to Implement:
 - Create a large K-W-L chart (Know–Want to Know–Learned) titled: “What do we know about blindness?”
 - Fill in the K and W columns as a group after reading Scene 1.
 - Leave space to return to the L column at the end of the unit or after the next lesson.

Let's Think!



Objective:

Students will understand how the structure and function of the human eye can inform the design of assistive technologies for people with visual impairments.

They will connect biological knowledge to real-world problem-solving.

Possible Answers:

1. Before Reading:

- “Learning how the eye works helps us understand what goes wrong when someone can’t see. Then we can think of tools or technology to help with those problems.”
- “If we know which part of the eye doesn’t work, we can make something that replaces or supports that function—like glasses, smart devices, or canes with sensors.”

2. After Reading (Reflection):

- “Now I understand that the retina sends messages to the brain. If that part is damaged, the brain won’t get the signal. We could make a device that sends signals in another way—like through sound or touch.”
- “I learned that the cornea and lens focus the light. Maybe we could create glasses that adjust focus automatically for people who have trouble seeing.”

Activity Link: <https://my.clevelandclinic.org/health/body/21823-eyes>



Possible Strategies for Implementation:

1. Think–Write–Pair–Share.

- How to Implement:
 - Students first think individually and write their answers.
 - Then they pair up with a classmate to compare what they thought before and after the reading.
 - Finally, ask a few students to share how their thinking changed after exploring the human eye website.

2. Interactive Web Exploration.

- How to Implement:
- Show the Cleveland Clinic website on a projector or allow students to explore it in pairs using tablets/laptops.
 - Pause at key diagrams and ask: “What do you notice about the eye’s parts?”
 - Encourage students to take short notes using labeled sketches.

3. Brainstorm Wall

- How to Implement:
 - Set up a large chart with columns: “Eye Function”, “What Could Go Wrong?”, and “Tech to Help”.
 - Let students contribute sticky notes based on what they learn from the website.
 - Use it to build class-wide ideas for future design challenges.

4. Visual Mapping

- How to Implement:
 - Have students draw a simple diagram of the eye and label each part’s job.
 - Ask them to write next to each label: “What if this part doesn’t work?”
 - This helps connect biology to invention thinking.

5. Mini Debate

- How to Implement:
 - Pose the question: “Is it more helpful to study the eye or the brain when designing for the blind?”
 - Let students choose a side and explain using evidence from the article.
 - This pushes critical thinking and evaluation skills.

Explorer’s Toolkit



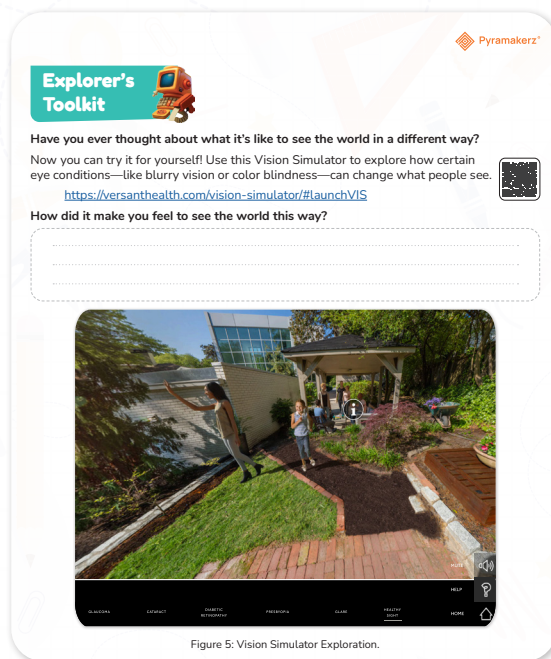
Teacher’s Role:

- Set up the activity by explaining the purpose of the Vision Simulator.
- Guide students through different vision conditions (e.g., blurry vision, color blindness).
- Facilitate observation and reflection by prompting students to compare what they see with normal vision.
- Encourage descriptive language and emotional responses.
- Support students in expressing how the changes might affect daily life.

Expected from students:

- Use the Vision Simulator tool to explore different types of visual impairments.
- Record or discuss what they noticed, felt, or were surprised by.
- Share observations with a partner or the class.
- Reflect on how vision changes could affect school, play, or movement.

Parts of the book included:



DIY Lab

Objective:

Students will simulate visual impairments using an online tool to develop empathy and firsthand understanding of how the world may look to someone with vision challenges.

Activity link: <https://versanthealth.com/vision-simulator/#launchVIS>

Activity Instructions:



1. Introduce the Simulation:

- Explain to students that they will explore what the world might look like to someone with a visual impairment using a special simulator.

2. Access the Vision Simulator:

- Direct students to the following link:
<https://versanthealth.com/vision-simulator/#launchVIS>
- Guide them to try out different vision conditions like:
 - Glaucoma
 - Cataract



- Diabetic Retinopathy
 - Presbyopia
 - Color Blindness
3. Observe Carefully:
- Ask students to describe what they see. What's different? What's difficult?
 - Encourage them to try moving around or interacting with the environment in the simulation (where possible).
4. Reflect in Writing:
- Have students respond to the question:
"How did it make you feel to see the world this way?"
 - Prompt them to think about emotions (e.g., frustrated, confused, nervous), physical reactions (e.g., squinting, leaning in), and practical challenges (e.g., reading signs, identifying people).
5. Class Discussion (Optional):
- Invite a few volunteers to share their thoughts.
 - Connect their insights back to the idea of inclusive technology and how designers must consider these experiences.

Possible answers:

How did it make you feel to see the world this way?

- "It made me feel confused and nervous because I couldn't see things clearly."
- "I didn't notice the people in the background at first. That would make it hard to know where to go."
- "It felt frustrating not being able to read the sign."
- "It helped me understand why tools like sound alerts or white canes are important."

Possible Strategies for Implementation:

- Strategy 1: Pair & Compare.
 - How to Implement:
 1. Have students work in pairs.
 2. Each student tries a different vision condition from the simulator.
 3. They then compare their experiences using sentence frames like:
 - "In my simulation, I couldn't see ____."
 - "It was easier/harder than I expected because ____."
- Strategy 2: Role Play Discussion.
 - How to Implement:
 1. After trying the simulation, assign each group a persona (e.g., student with glaucoma, someone with diabetic retinopathy).

2. Ask them to speak “in role” about what daily challenges they face.
 3. Use this to guide the class into designing better tech tools or accessibility features later.
- Strategy 3: Empathy Circle.
 - How to Implement:
 1. Set up a reflective circle where each student shares one word about how they felt during the simulation.
 2. After everyone shares, discuss why empathy is important in technology design—especially for visual impairments.

Explain



Teacher Role:

- Spark curiosity about the human eye and guide students to explore scientific explanations behind vision.
- Support vocabulary development (e.g., retina, lens, pupil) by modeling language and visuals during discussion.
- Facilitate student collaboration and peer dialogue after video viewing.
- Encourage thoughtful observation and questioning using the guiding prompt.
- Introduce the importance of understanding global health issues, especially vision-related challenges.
- Support students in reading and interpreting pie charts and infographics.
- Clarify difficult terms like “cataract,” “macular degeneration,” or “glaucoma” using simple definitions.
- Encourage students to reflect on how such conditions affect people’s daily lives and inspire empathy and innovation.

Expected From Students:

- Actively watch the video to understand how the human eye works.
- Engage in group or partner discussion about what they see and understand.
- Use their own words to explain how light travels through the eye and how different parts function.
- Write reflective responses based on video and image analysis.
- Read and interpret data from the infographic about global causes of blindness.

- Explore the linked article for more information about vision loss.
- Make connections between data and real-world problems.
- Begin thinking about solutions that support people with visual impairments.

Parts of the book included:


Screen Lessons

Click the link to learn more about how the eye works and what each part does.

<https://www.youtube.com/watch?v=axpCN6Vj9p0&pp=ygUVZXllIGFuYXRvbXkrlGZv>

After watching the video, take a close look at this picture.



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Dig Deeper

Did you know that millions of people around the world are blind or have trouble seeing?

This picture shows some of the main causes of vision loss.

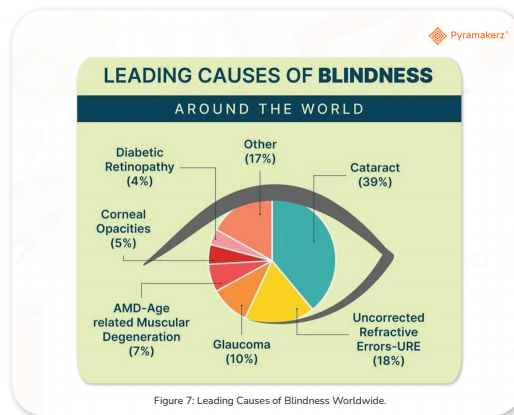
Read this article to learn more:

<https://my.clevelandclinic.org/health/diseases/24446-blindness>





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Screen Lessons

Objectives:

Students will be able to identify and describe the basic parts of the eye and explain how the eye processes light to help us see.

Activity Link: <https://www.youtube.com/watch?v=axpCN6Vj9p0&pp=ygUVZXllIGFuYXRvbXkrlGZv>



Activity Instructions:

1. Introduce the Topic:
 - Ask: “Have you ever wondered how we see things? What do you think happens inside your eye when you look at something?”
 - Show a picture of the human eye (or diagram if available).
2. Watch the Video:
 - Play the video linked above that explains the anatomy and function of the human eye.
 - Encourage students to take note of terms like retina, cornea, lens, and pupil.
3. Observe the Image (if one is shown after video):
 - Guide students to focus on what’s happening in the diagram/picture provided.
 - Use prompt: “What’s going on here?” to initiate observation and discussion.

4. Talk and Reflect:

- In pairs, have students talk about what the image shows and how it relates to the video.
- Then, ask them to write down their thoughts in the space provided.

Possible Student Answers:

- “Light goes through the pupil, then the lens focuses the light onto the retina.”
- “The retina sends messages to the brain so we can understand what we’re seeing.”
- “I see that the light enters the eye and bends as it goes through the lens.”
- “It shows that all the parts of the eye have to work together to help us see.”

Possible Strategies for Implementation:

- Visual Storytelling: “Eye Journey”
 - Have students draw the path of light as it enters the eye and label each part.
 - Use arrows and captions to tell the “story” of how we see.
- Think-Pair-Share
 - After video viewing, use the image prompt to:
 1. Think: Each student reflects independently.
 2. Pair: Students discuss ideas in pairs.
 3. Share: A few pairs share their explanation with the class.
- Concept Matching
 - Provide labels (pupil, retina, lens, cornea, optic nerve).
 - Ask students to match each label to the function they heard in the video or saw in the diagram.
- Interactive Whiteboard
 - Project the diagram and ask students to come up and label or describe what’s happening live, reinforcing vocabulary and understanding through interaction.



Dig Deeper

Objective:

Students will analyse visual data to identify leading causes of blindness worldwide and understand their global impact, building awareness and empathy that can inspire design solutions.

Activity Link: <https://my.clevelandclinic.org/health/diseases/24446-blindness>



Activity Instructions:

1. Introduce the Data Visualization:
 - Display the pie chart and ask, “What do you think this chart shows? Why do you think an eye is used in the design?”

2. Explore the Chart:

- Read aloud each cause and its percentage.
- Prompt: “Which cause is the biggest? Which ones surprised you?”

3. Read the Linked Article:

- Students can work in pairs or small groups to visit the article and read the overview on causes of blindness.
- Encourage them to pick one cause to learn more about.

4. Reflect & Discuss:

- Ask: “Why do you think it’s important for engineers and inventors to understand these causes?”
- Optional: Students create a short summary or infographic of one cause they found interesting.

Possible Strategies for Implementation:

1. Data Walk and Gallery Talk:

- Print and post the pie chart and additional infographics (if available) around the room.
- Students walk in small groups, read, and discuss each section using guiding questions.

2. Cause & Effect Web:

- On the board, list each cause of blindness. Ask students to draw connections:
 - Cause → How it affects vision
 - Cause → Possible assistive technologies

3. Pick and Present:

- Each student picks one cause, researches a little more (via article or teacher-provided materials), then presents:
 - What it is
 - How it affects people
 - One solution or helpful invention for that cause

4. Design Journal Prompt:

- “If you could design something to help someone with one of these conditions, what would you create? Why?”

Elaborate and Evaluate



Teacher Role:

- Introduce the Apprentice quiz as a fun review game: “Let’s test how much you’ve learned about the human eye!”

- Share the game link and explain how to look at clues and images to guess the correct eye part.
- Launch the Builder activity by showing the model-building video and highlighting how it demonstrates light passing through the pupil to the retina.
- Provide materials and monitor student progress as they build the model, asking guiding questions like, “Where does light go after it enters the eye?”
- Introduce the Mastermind activity as a take-home reflection: “Let’s think about how people around the world access eye care. Talk to someone at home and hear their perspective.”
- Explain that not everyone has equal access to eye doctors or glasses, and connect this to empathy and global understanding.
- Guide the Evaluate color blindness activity by encouraging students to walk around and find objects of different colors.
- Ask prompting questions like: “What would happen if those colors looked the same?” and “How might that affect someone’s daily life?”
- Lead a short discussion on the Ishihara test image and what it’s like for people with color blindness to view the world differently.
- Wrap up by helping students reflect on what they’ve learned and how understanding the eye can lead to real-world solutions.

Expected from Students

- Play the online quiz game and apply their knowledge of the eye’s anatomy to solve clues and identify parts.
- Watch the model-making video and construct a paper model showing how images form on the retina.
- Label the model accurately and use a flashlight to test how light travels through the eye structure.
- Ask a family member about access to eye care and write or draw their reflections on the differences between cities, villages, or countries.
- Observe and record the differences in identifying red and green objects and imagine life with color blindness.
- Try the color vision test by spotting the number in the Ishihara image and compare with classmates.
- Reflect on how understanding vision and its challenges can help in designing solutions for people with vision impairments.



Apprentice

Objective:

Students will review and apply their understanding of human eye anatomy by identifying and labelling key eye parts based on visual clues and descriptions.

Activity Link: <https://www.geoguessr.com/vgp/3802>



Possible Answers (Examples)

Students may identify:

- Cornea – clear outer layer protecting the eye.
- Iris – colored part controlling light entry.
- Pupil – dark center where light passes through.
- Retina – inner layer that senses light.
- Optic Nerve – sends signals to the brain.
- Lens – focuses light onto the retina.

Activity Instructions:

1. Introduce the Challenge.

- Tell students: “Today, you’ll test how well you know the parts of the eye! You’ll play a game where clues and images will help you figure out what part of the eye is being shown.”

2. Access the Link

- Students open <https://www.geoguessr.com/vgp/3802> on their devices.



3. Play the Game

- They look at each image and read the clue. Then, they select which part of the eye is being described.

4. Record Your Score

- Ask students to note how many they got correct and reflect on any questions they found tricky.

5. Pair & Share (Optional).

- Students can partner up to discuss the parts they struggled with and help each other review.

Possible Strategies for Implementation

1. Think-Pair-Share:

- Before starting, let students brainstorm eye parts they remember. After the game, discuss as pairs and share with the class.

2. Interactive Whiteboard Review:

- Display selected quiz items on the board. Students explain their thinking out loud before choosing an answer as a group.

3. Eye Diagram Recall:

- Provide a blank eye diagram before the game. Ask students to label it from memory, then compare after playing the game.

4. Game Show Format:

- Use a few sample clues in a “quiz show” style to energize the classroom before students go off on their own.



Objective:

Students will build a simple eye model to understand how light passes through the pupil and forms an image on the retina. This hands-on activity helps them visualize how the human eye processes images.



Activity Link: <http://youtube.com/watch?v=eL9JdQ-NaP8>

Materials Needed:

- A flashlight or phone light
- Black paper/card for the “pupil”
- White paper/card for the “retina”
- Scissors
- Tape or glue
- A printed or cutout object image (e.g., an arrow or symbol)
- Popsicle sticks or straws for stands

Activity Instructions:

1. Cut Out the Eye Parts:
 - Cut a small circular hole in the black paper to represent the pupil.
 - Keep a white circular paper behind it to act as the retina.
2. Create the Stand:
 - Tape the pupil and retina onto sticks or straws so they can stand vertically on a table.
3. Set Up the Light Source:
 - Place the object (e.g., arrow) in front of the pupil.
 - Shine the flashlight directly through the pupil onto the retina.
4. Observe the Image Formation:
 - Ask students to observe where the image appears.
 - Highlight how light travels in a straight line and how the pupil controls the light.
5. Discuss:
 - "What happened to the image on the retina?"
 - "Why do you think the image is upside-down?"
 - "How does our brain help us see it correctly?"

Possible Strategies for Implementation:

1. Model First, Then Create:

- Begin by showing the example video to clarify expectations.
- Use teacher’s model as a reference during student building time.

2. Think Aloud Strategy:

- While building the model, narrate each part aloud: “This small hole lets in light, just like the pupil...”

3. Partner Build & Explain:

- Pair students to build the model together and explain the parts to each other using sentence starters like:
 - “This is the pupil. It helps by...”
 - “The retina catches the...”

4. Gallery Walk Reflection:

- Set up a display of finished models and have students walk around to observe others' designs.
- Prompt questions like:
 - “What’s similar in all models?”
 - “What worked well for making the image clear?”



Mastermind

Objective:

Students will explore real-world challenges related to eye care access and color blindness. They will investigate and reflect on differences in healthcare accessibility and visual perception through inquiry-based, at-home tasks.

Activity Instructions:

1. Access to Eye Care (Mastermind Activity):

- In class, introduce the idea that not everyone has equal access to eye care (e.g., remote villages vs. cities).
- Assign students the task to interview a family member with this question:

“Do you think it’s easy for everyone in the world to get glasses or go to an eye doctor? Why or why not?”

- Then, ask them to write or draw their ideas about what could be different in a village, city, or country far away from them.
- Have them bring their completed responses next class to discuss as a group.

2. Color Blindness Task:

- Briefly explain what color blindness is and how it can affect daily tasks like reading signs or choosing clothes.
- Ask students to go home and find 3 red and 3 green items.
- Then, prompt them to imagine if those colors looked the same—how would that feel? What would be difficult to do?
- Lastly, have them look at the Ishihara test image (the dotted image with a number) and try to spot the number. Write or draw what they notice.

Possible Strategies for Implementation:

1. In-Class Launch Strategy:

- Use storytelling or real-life videos about people with color blindness or remote communities to spark empathy and curiosity.
- Display visuals of an eye clinic in a city vs. rural area or images showing how the world looks to someone with color blindness.
- Let students briefly share what they already know about glasses, eye care, or vision challenges to activate prior knowledge.

2. Scaffolding for Success:

- Give sentence starters or drawing prompts for students who may need help responding at home.
- Provide a printed version of the interview question and visual chart with red/green items for easy reference at home.
- Encourage families to talk together—builds community awareness and emotional connection.

3. Follow-Up Next Class:

- Create a “Wall of Insight” where students post their ideas and drawings.
- Facilitate a group reflection where students share what they found surprising or meaningful.



Now I Can

Objective:

Students will reflect on their learning journey in vision and eye health by identifying and expressing what they now understand about how the human eye works, the causes of vision loss, and the challenges people with visual impairments face. Through self-assessment using a "Now I Can" checklist, students will demonstrate their ability to apply new knowledge, empathize with others, and propose thoughtful solutions using the Engineering Design Process.

Activity Instructions:

1. Review Learning Targets:

- Display or read aloud the “Now I Can” checklist.
- Briefly revisit key moments from the lesson that align with each point (e.g., video about the eye, color blindness simulator, access to care reflection, EDP activity, etc.).

2. Student Reflection:

- Ask students to check off the statements they feel confident about.
- Encourage them to underline or circle one area they want to learn more about or still find challenging.
- Allow students to turn and talk to a partner about one “Now I Can” statement they feel proud of.

3. Optional Extension:

- Let students write or draw one thing they learned and how it might help them or someone else in real life.
- Use these reflections as an exit ticket or assessment of understanding.

Relation to Standards.

1. Next Generation Science Standards (NGSS):

- **LS1.A: Structure and Function** – Students explore the structure of the human eye and understand how its parts work together to perform the function of sight.
- **ETS1.A: Defining and Delimiting Engineering Problems** – Students identify real-world problems faced by people with vision loss and begin developing solutions using the Engineering Design Process.

2. ISTE Standards for Students:

- **Empowered Learner** – Students use digital tools (videos, simulations, online games) to explore how the eye functions and reflect on their learning.
- **Innovative Designer** – Students apply design thinking to explore assistive technologies and brainstorm ways to help those with visual impairments.

3. United Nations Sustainable Development Goals (SDGs):

- **SDG 3: Good Health and Well-Being** – Students learn about global eye health challenges and discuss solutions for vision care accessibility.
- **SDG 10: Reduced Inequalities** – Students reflect on unequal access to healthcare in different regions and consider inclusive design solutions.